

Unsupervised domain adaptation for SAR target classification based on domain- and class-level alignment: From simulated to real data[☆]

Yu Shi^a, Lan Du^{a,*}, Chen Li^a, Yuchen Guo^b, Yuang Du^a

^a National Key Laboratory of Radar Signal Processing, Xidian University, Xi'an, 710071, China

^b Academy of Advanced Interdisciplinary Research, Xidian University, Xi'an 710071, China

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ABSTRACT

Synthetic Aperture Radar (SAR) target classification methods based on convolutional neural networks have attracted wide attention recently. Typically, these methods necessitate a substantial quantity of labeled data to train the deep model, which is an obstacle for SAR target classification since annotating SAR images is costly and time-consuming. Generating large-scale labeled data in simulation to train the deep model is a feasible alternative. However, cross-domain recognition is challenging due to significant differences in distribution between simulated and real domains. In this paper, we propose an unsupervised domain adaptation method for SAR target classification by aligning the distribution between the simulated and unlabeled real domains. Specifically, we encode the dependencies across different granularity perspectives including domain-, and class-levels simultaneously to align two domains finely. On the one hand, a gradient-weighted adversarial alignment method is proposed to align the global distributions at the domain-level. On the other hand, a novel class-level alignment method based on the prototypical network is proposed to align the finer class structure. Technically, we respectively model the cross-domain prototype-prototype relations and cross-domain prototype-instance relations with contrastive learning to achieve intra-class prototype compact as well as inter-class prototype separation among cross-domain categories, where the prototypes of the real domain are learned with pseudo labels. To improve the quality of pseudo-labels, a pseudo-label filtering strategy combined with the attribute scattering center is further designed. The experiments on two simulation-to-reality datasets indicate that the proposed method outperforms state-of-the-art unsupervised domain adaptive methods, such as pixelfeat (+4.45% and +7.6% in accuracy on SAMPLE and S2M datasets respectively). Code and data will be available in <https://github.com/YuShi1213/Sim2Real-Unsupervised-SAR-Target-Classification>.

1. Introduction

As an active microwave detection sensor, synthetic aperture radar (SAR) has become one of the main remote sensing detection systems due to its technical advantages such as all-weather, all-day, and long-range, and has been playing a very important role in military and civilian fields since its birth (Moreira et al., 2013). With the gradual maturity of SAR systems and imaging algorithms, more and more high-resolution and high-quality SAR images appear. More information can be obtained by using SAR imaging technology, which provides strong technical support for high-definition reconnaissance and surveillance of targets. Among these applications, SAR target classification has attracted wide attention due to its essential value in military surveillance, urban planning, ocean monitoring, and so on (Wen et al., 2021).

Recently, significant success has been achieved with SAR target classification methods that employ convolutional neural networks (CNNs) (Zhu et al., 2021) since CNNs can automatically learn effective hierarchical features in a manner of data-driven. Some typical SAR target classification methods based on CNNs are proposed, including 1) traditional CNNs-based methods, such as A-ConvNets (Chen et al., 2016), DCNN-SAR (Morgan, 2015), all-in-one CNN (Ding et al., 2016), SNN (Chen et al., 2022a), IDCNN (Gao et al., 2019) MMFF (Liu and Lang, 2021), 2) complex-valued CNNs, such as deep SAR-Net (Huang et al., 2020), and 3) CNNs combined attribute scattering center, such as CA-MCNN (Li et al., 2021), ASC-MBRCN (Feng et al., 2022). However, these methods necessitate a substantial quantity of labeled data to train the deep model, which is an obstacle for SAR target classification since annotating SAR images is costly and time-consuming, especially for

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* Corresponding author.

E-mail addresses: yshi_1@stu.xidian.edu.cn (Y. Shi), duan@mail.xidian.edu.cn (L. Du), chenli@stu.xidian.edu.cn (C. Li), yghuo@xidian.edu.cn (Y. Guo), yadu_1@stu.xidian.edu.cn (Y. Du).

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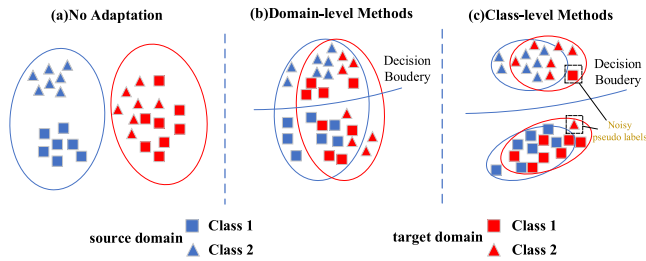


Fig. 1. Motivation of the proposed method. (a) The source and target domains have a large domain discrepancy without domain adaptation. (b) Domain-level alignment methods align the global distributions between domains and do not take into account the class information, which may lead to some samples of different classes being aligned incorrectly. Besides that, the decision boundary learned by global alignment might generalize poorly for the target domain. (c) Class-level methods use the pseudo labels of the target domain to guide class-level alignment. However, the performance of these methods is generally related to the quality of the generated pseudo-labels.

data with fine-grained categories (Zhu et al., 2021). Generally, for ordinary people, it is difficult to understand and interpret SAR images, and professional expertise is required for labeling SAR images (Zhao et al., 2023). Consequently, the acquisition of large-amount labeled data is difficult for training a deep SAR target classification model. Training the deep model in simulation is becoming a feasible alternative. Despite the efforts to build good dynamic models and high-fidelity simulators for scalable data collection, the modeling inaccuracies still lead to significant differences in distribution between simulated and real domains (Balz et al., 2015), making the deep target classification model trained in simulation difficult to generalize to the real world (Cha et al., 2018), i.e., cross-domain recognition challenging.

To tackle the above challenge, unsupervised domain adaptation (UDA) emerged as a solution to align the distributions by minimizing the disparity between the labeled source and unlabeled target domain. At present, some UDA methods have been applied to SAR target classification fields. Most of the effective methods rely on aligning the global distributions using maximum mean discrepancy (MMD) (Huang et al., 2019), Joint MMD (Geng et al., 2020), or adversarial learning (Xiaoling et al., 2022). However, these previous works only align the distributions at the domain-level, neglecting the finer class distribution structure (class-level) between two domains. For an target classification task, it is essential to align the category structure. As shown in Fig. 1, the affinity of various classes across the two domains towards alignment might vary, which may lead to some samples of different classes being aligned incorrectly. Besides that, the decision boundary learned by global alignment might generalize poorly for the target domain. Many subsequent works of the computer vision field, such as PFAN (Chen et al., 2019), DSAN (Zhu et al., 2020), ATDOC (Liang et al., 2021), TPN (Pan et al., 2019), CAN (Kang et al., 2019), ILA-DA (Sharma et al., 2021), SRDC (Tang et al., 2020), have alleviated this problem to some extent by using pseudo-labels of the target domain to guide class-level alignment. They initially assign target examples a pseudo label by feature clustering or source classifiers. They then align the distributions of the same classes among domains and update the pseudo-labels with self-training manner. However, in most cases, the performance of these methods is related to the quality of the predicted pseudo-labels, which may be noisy and lead to negative alignment between different classes if there are not sufficient filtering measures. In computer vision, they use a pre-training model with the super large-scale ImageNet (Deng et al., 2009) to obtain a generic feature extractor and then directly predict pseudo labels by feature clustering or source classifiers, which can guarantee the reliability of target pseudo-labels to some extent. Some co-teaching-based methods utilize dual classifier head that is assisted by curriculum learning to obtain more reliable pseudo-labels (Saha et al., 2022). Based on the co-teaching with pseudo-labels, they learn more discriminative target representations and achieve class-level alignment.

However, since optical and SAR images have dramatic differences in imaging mechanisms, pseudo-label filtering strategies designed for SAR data should be considered.

Building on this line of research, our objective is to address the challenges associated with unsupervised SAR target classification by aligning the distributions between the unlabeled real SAR domain and the labeled simulated domain. The primary issues that still require resolution can be summarized as follows:

- From a theoretical point of view, not only the domain-level alignment but also the class-level alignment should be considered for the UDA SAR target classification method.
- From an application point of view, an effective pseudo-label filtering strategy for SAR data should be proposed to ensure the correct guidelines of class-level alignment.

To solve this two challenges, we propose a novel UDA method for SAR target classification by aligning the distribution between simulated and unlabeled real SAR domains across different granularity perspectives including domain-, and class-levels simultaneously. For domain-level alignment, a gradient-weighted adversarial learning method is proposed to align the global distributions between domains at the domain-level. Compared to the traditional adversarial learning methods, for further aligning hard-confused samples, we down-weight the gradients of easily confused samples and up-weight the gradients of hard confused samples according to the confusion of samples to align the global distributions better. For class-level alignment, a novel method based on the prototypical network is proposed to align the finer class structure. The prototypical network (Movshovitz-Attias et al., 2017), which can better encode the semantic class structure of data, assumes that each class in the embedding space can be represented by a prototype (or centroid), and the classification result can then be obtained by calculating the distance between the instance and the prototype of each class. In this way, the class-level adaptation is represented as intra-class prototype compact as well as inter-class prototype separation among cross-domain categories. To achieve this goal, we respectively model the cross-domain prototype-prototype relations and cross-domain prototype-instance relations with contrastive learning, where the prototypes of the real domain are learned with pseudo-labels. The pseudo-labels are gradually updated with a self-training manner. However, some predicted pseudo-labels by the deep model are unreliable. The attribute scattering center can reflect the target's physical structure, which can supplement prediction results of the deep model with helpful information (Li et al., 2021). To ensure the correct guidelines of the class-level alignment, a pseudo-label filtering strategy combined with the attribute scattering center is further designed. More specifically, our main contributions can be described as follows:

1. **To solve the problem of lacking labeled data for training the SAR target classification network, a novel UDA method is proposed.** Proposed method aligns the distribution between simulated and unlabeled real SAR domains across different granularity perspectives including domain-, and class-levels simultaneously. To the best of our knowledge, it is the first work to introduce the idea of class-level alignment into the SAR target classification task.
2. **A novel class-level alignment method based on the prototypical network is proposed to align the finer class structure.** The cross-domain prototype-prototype relations and cross-domain prototype-instance relations are respectively modeled with contrastive learning to achieve intra-class prototype compact as well as inter-class prototype separation among cross-domain categories.
3. **A gradient-weighted adversarial learning method is proposed to align the global distributions at the domain-level.** For further aligning hard-confused samples, we down-weight the gradients of easily confused samples and up-weight the gradients of hard confused samples according to the confusion of samples to align global distributions better.

4. To solve the problem of severe unreliability of initial pseudo-labels in the field of SAR target classification by class alignment method, a noisy pseudo-label filtering strategy combined with the attribute scattering center is proposed. The proposed strategy effectively improves the reliability of the initial pseudo-labels.

Next, the related work will be discussed in Section 2, the preliminary knowledge will be introduced in Section 3, Section 4 will introduce the proposed UDA method for SAR target classification based on multi-granularity alignment. Section 5 will present the experiments and analysis. Finally, Section 6 will give the conclusion.

2. Related work

In this section, some methods of SAR target classification with domain adaptation methods will be first presented and then we will follow some domain adaptation methods of remote sensing data and finally, several related methods of class-level-based unsupervised domain adaptation methods will be introduced.

2.1. Domain adaptive SAR target classification

Despite the impressive performance of deep learning based SAR target classification methods, they generally require a substantial quantity of labeled data to train the SAR target classification model. Due to the characteristics of SAR images, annotating the SAR data is costly and time-consuming. As a tackle solution, the domain adaptation approach is introduced in SAR target classification. Some researchers have explored using the labeled source domain to assist the target classification in the less-labeled SAR domain. Huang et al. (2019) explored adapting with maximum mean discrepancy (MMD) from multiple domains to the SAR domain. Wang et al. (2019) introduced a method that uses simulated data to alleviate the issue of limited real SAR samples. They accomplished this by combining two techniques: meta-learning and adversarial alignment. He et al. (2021) proposed a SAR target classification method that used the MMD to align the distributions between simulated and real domains to improve performance. Bi et al. (2023) proposed a pixel-level-based few-shot domain adaptation method that reduces the difference in the distribution and obtains the reconstructed simulated SAR images for few-shot target classification. The above methods still need labeled SAR data to train the network. To further reduce the cost of labeling, some researchers have explored unsupervised domain adaptation for SAR target classification. Xiaoling et al. (2022) align the global distribution between the simulated domain and unlabeled real domain using MMD and adversarial learning. Zhao et al. (2022) proposed an unsupervised domain adaptation method under open set condition, which achieved promising performance by aligning the distribution of SAR domains crossing different satellites with adversarial learning and a pseudo-labeling strategy. Chen et al. (2022b) proposed a two-stage UDA method with adversarial learning. They first generated images from simulated images to real images via CycleGAN Zhu et al. (2017) and then aligned the feature distribution with adversarial learning between the generated real images and real images. However, since GAN networks are difficult to train, this method is susceptible to the quality of the generated images. Geng et al. (2020) proposed a deep adaptive method to align the distribution by extracting features of different SAR domains into a shared subspace. Although existing domain-adaptive SAR target classification methods have made some progress, these works only aligned the global distributions at the domain-level, neglecting the finer class distribution (class level) alignment between two domains, which may lead to some samples of different classes being aligned incorrectly. In this paper, we propose a novel UDA method for SAR target classification, which aligns the distribution between the simulated and unlabeled real domains at domain-, and class-levels simultaneously, achieving better performance.

2.2. Domain adaptation for remote sensing data

There are many domain adaptation methods designed for remote sensing data, which are not limited to target classification tasks. They can be roughly divided into generative-based methods, adversarial-based methods, pseudo-label-based methods, and hybrid methods. Generative-based methods aim to generate images that are similar in appearance to the target domain in order to use labels from the source domain to develop a model that can also generalize to the target domain. A representative method is the segmentation method proposed by Benjdira et al. (2019) in the ISPRS dataset, which first uses GAN to generate the target domain images and then fine-tunes it. The adversarial-based methods aim to extract domain-invariant features of the source domain and target domain so that the labeled source domain features guide the learning of the target domain. A representative method is the cross-domain road detection method proposed by Lu et al. (2021b) which learns cross-domain spatial information consistency through global adversarial alignment and learns local semantic consistency through local adversarial alignment. Rostami et al. (2019) proposed a domain adaptive ship classification method that reduces the discrepancy between the optical and SAR data. Pseudo-label-based methods aim to learn the discriminative features of the target domain by using pseudo-labels, which generally train the model in a self-training manner. GraphDA (Saha et al., 2022) proposes a multi-target domain adaptation method for remote sensing image classification, which uses dual classifier heads to assign pseudo-labels, one of which is a graph convolutional network (GCN). To prevent the classifier from overfitting its own noisy pseudo-labels, the method develops a co-teaching strategy of dual classifier heads assisted by curriculum learning to obtain more reliable pseudo-labels. Hybrid methods are usually a combination of the above methods and occupy the mainstream of existing domain adaptation methods. To overcome the shortage of labeled SAR images for finetuning deep models, Ye et al. (2019) proposed an unsupervised domain adaptation for SAR image retrieval with the help of the optical dataset, which built a novel adversarial network by adding the domain discriminator and pseudo-labels predictor. To solve the problem of lacking labeled SAR data to train the ship detector, Shi et al. (2022) proposed an unsupervised domain adaptation method for SAR ship detection with the help of labeled optical images, which progressively transfers the knowledge from the pixel-level, feature level, and prediction level. To solve the problem of distribution mismatch between the source radar and target radar data, Shi et al. (2021) generate target images via CycleGAN to alleviate the problem of limited SAR images and learn useful features of SAR images by feature alignment and discriminative learning, which can transfer knowledge from one SAR domain to another SAR domain.

2.3. Class-level unsupervised domain adaptation

Many subsequent works in the computer vision field took account into class distribution structure between domains, achieving better performance. Since class labels for the target domain are not available, most existing methods rely on utilizing pseudo-labels to perform class-level alignment. Some researchers explored instance-level cross-domain contrastive learning. Kang et al. (2019) introduce contrastive learning into domain adaptation to align the finer class distributions with improved MMD. They adopted the spherical K-means to assign pseudo labels of the target domain. CoSCA (Dai et al., 2020) uses the triplet loss Hermans et al. (2017) to achieve cross-domain contrastive learning. Sharma et al. (2021) suggested a technique called ILA-DA, which employs a multi-sample contrastive loss function to facilitate the process of aligning domains, achieving a more compact representation of classes and also increasing the separation between classes across different domains. These methods introduce contrastive learning into class-level alignment, achieving promising performance.

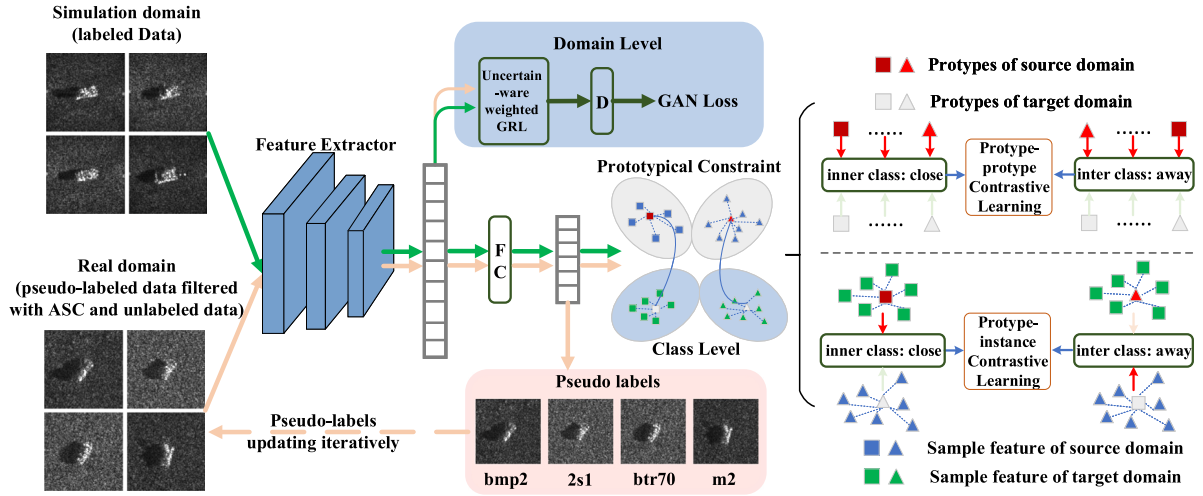


Fig. 2. The overall architecture of the proposed UDA method based on domain- and class-levels alignment for SAR target classification, where D represents the domain discriminator, and FC represents the fully-connected layer. Based on the prototype network, our method aligns the distribution between simulated and unlabeled real SAR domains across different granularity perspectives including domain-, and class-levels simultaneously. For domain-level alignment, we propose a gradient-weighted adversarial learning method to align the global distribution between domains. For class-level alignment, based on the pseudo-labels filtered with ASC strategy, we model the cross-domain prototype-prototype relations and cross-domain prototype-instance relations with contrastive learning to achieve intra-class prototype compact as well as inter-class prototype separation among cross-domain categories. In the training process, the pseudo-labels are updated iteratively with a self-training manner.

However, they depend on pairs of data points, empirically suffering from sampling issues of positive/negative sample pairs. Besides that, the above methods lack effective pseudo-label filtering solutions. To obtain less noisy pseudo labels, Liang et al. (2021) considered the target class structure to refine the original labels. To reduce the sampling issues, some researchers explored the prototype-based UDA methods. Pan et al. (2019) align the cross-domain inner-class prototypes between the source, source-target, and target domains with pseudo-labels. Chen et al. (2019) reduce the domain shift by aligning the category prototype between two domains. Besides that, they proposed an easy-to-hard transfer strategy to alleviate the negative effect of wrong pseudo-labels. Technically, they set an adaptive threshold, and only pseudo-labeled samples exceeding this threshold are selected for training. Although the idea of alignment by class is introduced, these works only model the compactness of the cross-domain inner-class and do not consider the separation of the cross-domain inter-class. In contrast, our method proposes a novel cross-domain prototype modeling mechanism by introducing contrastive learning into class-level domain adaptation, which achieve intra-class prototypes compact as well as inter-class prototypes separation among cross-domain categories. To alleviate the negative effect of wrong pseudo-labels, we further propose a noisy pseudo-label filtering strategy combined with the attribute scattering center to improve the quality of target pseudo-labels.

3. Preliminary knowledge

The preliminary knowledge of our method will be introduced in this section to better understand our method, including the prototypical network and self-training for UDA.

3.1. Overview of the prototypical network

The prototypical network assumes that each class of the data points (samples) can be characterized by a prototype (or centroid) in the embedding space and the classification result then can be obtained by calculating the distance between the data point and each category's prototype (Movshovitz-Attias et al., 2017; Snell et al., 2017; Lu et al., 2022). Formally, given a set of N labeled examples with c classes $S = \{(v_i, y_i)\}_{i=1}^N$, where each $v_i = G(x_i) \in \mathbb{R}^D$ is the D -dimension feature vector extracted by the feature extractor G and $y_i \in \{1, 2, \dots, C\}$ is the label. Through an embedding function $f_\phi : \mathbb{R}^D \rightarrow \mathbb{R}^M$, the

prototypical network computes an M -dimensional representation of each class $\mu_c \in \mathbb{R}^M$, named prototype. Each prototype is a group of learnable parameters $\mu_c(\theta) \in \mathbb{R}^M$ (Movshovitz-Attias et al., 2017) or the mean vector of the embedded features belonging to its category $\mu_c = \frac{1}{|S_c|} \sum_{(v_i, y_i) \in S_c} f_\phi(v_i)$ (Snell et al., 2017). In our method, we use the $\mu_c(\theta) \in \mathbb{R}^M$ as the prototype.

Given a distance function $d : \mathbb{R}^M \times \mathbb{R}^M \rightarrow [0, +\infty)$, for an instance point v , the prototypical network produces a distribution score $P(y = c|v)$ over c classes based on a softmax over distances to the prototypes:

$$P(y = c|v) = \frac{\exp(-d(f_\phi(v), \mu_c))}{\sum_{c'} \exp(-d(f_\phi(v), \mu_{c'}))}, \quad (1)$$

The training process involves minimizing the negative log-likelihood probability of assigning the correct class label c to a given sample:

$$\mathcal{L}_{\text{supervise}} = -\log P_\phi(y = c|v). \quad (2)$$

3.2. Self-training for UDA

In the setting of UDA, the data available in the target domain is devoid of any labeling information or annotations. However, class-level alignment-based domain adaptation methods rely on the guidance of annotation information. Self-training (Shi et al., 2022; Zoph et al., 2020) is a widely used pseudo-label learning method.

Self-training divides the training process into alternate two steps: 1) Fixing the parameters of the network and updating the pseudo-labels; 2) Based on the updated labels, updating the parameters of the network through back-propagation. These two steps are alternated in each training iteration or epoch. Most class-level alignment UDA methods in computer vision perform self-training for pseudo-label learning. The effectiveness of self-training is highly reliant on the accuracy and reliability of the initial pseudo-labels that are generated to train the model. If the initial pseudo-labels are of poor quality, the self-training process will likely result in a model with suboptimal performance. Therefore, it is crucial to ensure that the initial labeling is as precise as possible to achieve the best results in self-training. If there is an effective noisy pseudo-label filtering strategy, self-training can progressively capture the structure of unlabeled samples and further refine/generate pseudo-labels for pseudo-labeled/unlabeled samples to learn better feature representations.

4. Proposed method

In this section, the details of the proposed UDA method based on domain-level and class-level alignment for SAR target classification will be introduced. Formally, in the task of UDA SAR target classification, given a set of fully labeled simulated SAR data (source domain) $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, where x_i^s and y_i^s denote the simulated data and corresponding class labels, and a set of unlabeled target real data (target domain) $D_t = \{(x_j^t)\}_{j=1}^{N_t}$, where x_j^t represents the unlabeled real SAR data. We follow the rules of inductive learning and divide the unlabeled real data into two subsets: training set and test set. We aim to learn a cross-domain classifier using labeled simulated SAR data that can perform well on the unlabeled real SAR data by aligning the distributions between the labeled simulated SAR and unlabeled real SAR domains. Fig. 2 gives the framework of the proposed UDA method for SAR target classification. Our method is based on the prototypical network, which aligns the distribution between simulated and unlabeled real SAR domains across different granularity perspectives. The proposed method consists of two main novel components. Firstly, it proposes a gradient-weighted adversarial alignment technique to align the global distributions at the domain-level. Secondly, the paper presents a class-level alignment method based on the prototypical network. Our method utilizes contrastive learning to model cross-domain prototype-prototype relations and prototype-instance relations. Pseudo labels are assigned and utilized during training to learn prototypes in the real domain. We follow the general way to pre-train the model with simulated SAR data and find pseudo-labels of real (unlabeled) SAR data with the pretrained model. To improve the quality of pseudo-labels, a novel pseudo-label filtering strategy combined with the attribute scattering center is designed.

Next, we will introduce the details of domain-level alignment, named gradient-weighted adversarial learning, and class-level alignment, named cross-domain prototype contrastive learning.

4.1. Gradient-weighted adversarial learning

Gradient-weighted adversarial learning (GVAL) aims to align the global distributions between the simulated and unlabeled real domains. As shown in Fig. 2, Given the input x , the feature extractor G extracts the corresponding lower dimensional feature representations $v_i = G(x_i) \in \mathbb{R}^D$, where the parameters of G are shared between the labeled simulated domain and the unlabeled real domain. The prototype classifier then produces a distribution over classes in the embedding space on the labeled source data. However, since the distributions between the simulated and real domains have significant differences, the prototype classifier that was trained using the simulated data does not exhibit good performance when it is applied to the real data. The adversarial learning strategy (Ganin and Lempitsky, 2015; Tzeng et al., 2017; Lu et al., 2021a) is proven to be effective for aligning global distributions. A domain discriminator D , essentially a binary classifier, aims to classify whether a feature vector v comes from the simulated or real SAR domain. The feature extractor aims to deceive the discriminator to generate confused features. An adversarial learning manner is used to train the feature extractor and discriminator. The loss is as follows:

$$\begin{aligned} \mathcal{L}_{dom_adv}(G, D, X_{simulate}, X_{real}) \\ = E_{x_{simulate} \sim X_{simulate}} [\log D(G(x_{simulate}))] \\ + E_{x_{real} \sim X_{real}} [\log(1 - D(G(x_{real})))] \end{aligned} \quad (3)$$

The goal of D is to minimize the objective and G is to maximize the objective. The G tends to learn the domain-confusion features and achieve the goal of global distribution alignment after this adversarial process. The gradient reversal layer (GRL) (Ganin and Lempitsky, 2015) is generally used to achieve the end-to-end adversarial training procedure and is proven effective. During forward propagation, GRL keeps

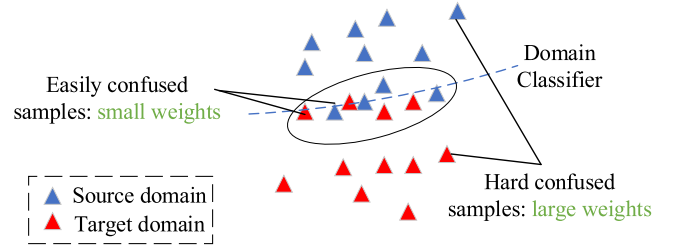


Fig. 3. Illustration of gradient weighted strategy. The blue triangles denote the features of the source samples and the red triangles denote the features of the target samples. The samples that make the domain classifier more confuse, i.e., close to the decision boundary, are considered as easily confused samples and should be down-weighted. The samples that make the domain classifier easily distinguishable, i.e., outside the decision boundary, are considered as hardly confused samples and should be up-weighted. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the features unchanged and during backpropagation, GRL reverses the gradient of the features between the D and the G :

$$grad_{rev} = -\lambda \cdot grad. \quad (4)$$

Where $grad$ and $grad_{rev}$ represent the gradient before and after reverse. The gradient of the feature extractor and domain classifier is reversed in the optimization process to ensure the adversarial process.

The traditional GRL treats each sample equally, ignoring that the difficulty of different samples for domain alignment is different. We propose a gradient-weighted strategy to improve GRL. As shown in Fig. 3, to achieve better domain confusion, we aim to improve GRL by down-weighting the easily-confused samples and up-weighting the hard-confused samples. Looking back at generative adversarial networks (GANs) (Goodfellow et al., 2020), when the generator produces a feature that makes the domain classifier more distinguished, that feature is more confusing, and vice versa. A typical example is when the domain classifier predicts a probability of 0.5 for a feature, i.e., most undistinguishable by the classifier, its level of confusion is highest (Goodfellow et al., 2020). Thus, we should let the network pay less attention to the easily confused samples by down-weighting their gradients. And we should let the network pay more attention to hardly confused samples by up-weighting their gradients. Specifically, the proposed gradient-weighted GRL leverages the confusion of the D as the weights for the corresponding features. Given an input x , the probability of sample x belonging to the source domain predicted by the D is p . Then, we use the entropy of the domain classifier's output to measure the confusion of each x :

$$u = (p \cdot \log \frac{1}{p} + (1-p) \log \frac{1}{(1-p)}), \quad (5)$$

Suppose the gradient before the reversal to be $grad$, and the gradient after the reversal to be $grad_{rev}$, then our method can be described as follows:

$$grad_{rev} = -\lambda \cdot grad \cdot (1 - u). \quad (6)$$

Our method down-weights the easy-confused samples and simultaneously up-weights the hard-confused samples by reweighting the gradients, achieving better domain confusion.

4.2. Cross-domain prototype contrastive learning

The prototype of a class can well represent the structure of the whole class. Therefore, we aim to reduce the domain gaps by modeling the relations of the cross-domain prototypes to achieve class-level alignment. However, in the setting of UDA, the real SAR domain is unlabeled and the target prototypes cannot be gained directly. The pseudo-labels of the target domain assigned by the source-trained model are also unreliable and may lead to the misalignment of the two

domains. To ensure the correct guidelines of the class-level alignment, a pseudo-label filtering strategy combined with the attribute scattering center is further proposed to improve the reliability of pseudo-labels effectively. Then, based on refined target pseudo-labels, for class-level alignment, we model the cross-domain prototype-prototype relations and cross-domain prototype-instance relations with contrastive learning to achieve intra-class prototype compact as well as inter-class prototype separation among cross-domain classes.

(1) Pseudo-label Filtering Strategy Combined with Attribute Scattering Center

As mentioned above, it is unreliable when directly utilizing the classifier learned on the labeled simulated data to assign the pseudo labels for real samples. The attribute scattering center can reflect the physical structure information of the target, which can supplement the prediction results of the deep model with helpful information. We aim to filter the unreliable pseudo labels by combining the ASC.

Generally, we can assign pseudo-labels for the target domain by matching each sample of the target domain with the closest source prototype. Assume the pseudo-label predicted by the source classifier for the target sample x_j^t as $\hat{y}_j^t(\text{source})$. As mentioned above, some predicted pseudo-labels by the deep model are unreliable. Motivated by Zhang (2019), ASC matching can also be used to recognize the target and can supplement the prediction results of the deep model. We view the source domain as the template set and view the target domain as the test set. Then we follow Zhang (2019) to extract the ASC based on sparse representation and use the Hungarian algorithm to match ASC for target classification. The four attributes $[A, x, y, L]$ are chosen since these four attributes depict the significant physical characteristics of the target, where A is the amplitude, x and y are coordinates in range and azimuth dimensions, L is the length. For the test ASC set (training samples of target domain) $P = [u_1, u_2, \dots, u_{N_s}]$ and templet ASC set (training samples of source domain) $w = [w_1, w_2, \dots, w_{N_t}]$, the distance between two ASCs can be calculated as:

$$d(u_i, w_j) = [(u_{i,x} - w_{j,x})^2 + (u_{i,y} - w_{j,y})^2 + (u_{i,L} - w_{j,L})^2 / 2] \cdot \exp((u_{i,A} - w_{j,A}) / 2). \quad (7)$$

Where $u_{i,x}, u_{i,y}, u_{i,L}, u_{i,A}$ represent four attributes of u_i , and similar for w_j .

The Hungarian algorithm, an efficient combinatorial optimization algorithm for solving task assignment problems, is performed to match the two ASC sets based on the distance. After ASC matching, we can also assign each target sample a pseudo label. Assume the pseudo label predicted by the ASC matching classifier for the target sample x_j^t as $\hat{y}_j^t(\text{ASC})$. For each target sample x_j^t , we can obtain two prediction results and we fuse these two predictions to filter the unreliable pseudo labels. Specifically, we only keep the pseudo-labels that the two classifiers predict to be consistent, and filter the pseudo-labels that predict inconsistently. The filtering strategy is shown in Algorithm 1.

After the pseudo-label filtering strategy, a refined target domain with pseudo-labels can be obtained and we can learn a better feature representation and corresponding target prototypes for the target domain. Note that, we only perform this strategy to filter the initial pseudo labels of the target domain since the accuracy of ASC matching will not improve. Then, we can perform self-training to progressively capture the target distributions and thus refine/generate pseudo-labels for pseudo-labeled/unlabeled real SAR data to learn better feature representations.

(2) Cross-domain Prototype-prototype and Prototype-instance modeling

Given a set of fully labeled simulated SAR data (source domain) $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, where x_i^s and y_i^s denote the simulated data and corresponding class labels, a set of pseudo-labeled real SAR data (pseudo target domain) $D_t = \{(x_j^t, \hat{y}_j^t)\}_{j=1}^P$, where x_j^t and $\hat{y}_j^t$ denote the real data

Algorithm 1 Pseudo-label filtering strategy combined with ASC

Input: Target SAR training samples; prototype-based classifier trained on source samples CLS_{PRO} ; ASC-matching-based classifier CLS_{ASC} .

Output: refined pseudo labels set R

- 1: Initialize R as an empty set;
- 2: for x_j^t in target domain do:
- 3: $\hat{y}_j^t(\text{source}) = CLS_{PRO}(x_j^t)$;
- 4: $\hat{y}_j^t(\text{ASC}) = CLS_{ASC}(x_j^t)$;
- 5: if $\hat{y}_j^t(\text{source}) = \hat{y}_j^t(\text{ASC})$:
- 6: Select $(x_j^t, \hat{y}_j^t)$ and add it into R .
- 7: return R .

and corresponding pseudo-class labels, and an unlabeled target domain $D_t = \{(x_j^t)\}_{j=M-P}^M$, we can learn two kinds of classifiers, namely prototypes of the source domain $\mu_c^s(\theta)$ and prototypes of the target domain $\mu_c^t(\theta)$. For class-level alignment, we model the cross-domain prototype-prototype relations and cross-domain prototype-instance relations with contrastive learning to achieve intra-class prototype compact as well as inter-class prototype separation among cross-domain classes. The class-level alignment with labeled samples can progressively learn the structure of target distributions and thus refine/generate pseudo labels for pseudo-labeled/unlabeled target samples to learn better feature representations with a self-training manner.

Our class-level alignment aims to achieve the following two goals:

Goal 1: We should pull the prototypes of the same classes across the source and target closer to each other while pushing away the prototypes of different classes.

Following contrastive loss is proposed to attain the goal:

$$\mathcal{L}(\mu_c^s(\theta), \mu_c^t(\theta)) = - \sum_{c=1}^k \log \frac{e^{d(\mu_c^s(\theta), \mu_c^t(\theta))}}{\sum_{c'} e^{d(\mu_c^s(\theta), \mu_{c'}^t(\theta))}}, \quad (8)$$

Where $d(\mu_c^s(\theta), \mu_c^t(\theta))$ represent the function to calculate the prototypes' distance between the source and target domain.

Goal 2: We should pull the source samples closer to the target prototype of the same class while pushing the source samples away to target prototypes of the different classes and vice versa.

Following contrastive loss is proposed to attain the goal:

$$\mathcal{L}(e, \mu(\theta)) = - \sum_{c=1}^k \left(\log \frac{e^{d(e_c^t, \mu_c^t(\theta))}}{\sum_{c'} e^{d(e_c^t, \mu_{c'}^t(\theta))}} + \log \frac{e^{d(e_c^t, \mu_c^s(\theta))}}{\sum_{c'} e^{d(e_c^t, \mu_{c'}^s(\theta))}} \right), \quad (9)$$

Where e_c^t represents the embedding vector of source samples belonging to class c , and $d(e, \mu(\theta))$ represents the function to calculate the distance between cross-domain prototypes and sample instances.

Then the objective of the class-level alignment can be written as:

$$\mathcal{L}_{class} = \mathcal{L}(\mu_c^s(\theta), \mu_c^t(\theta)) + \mathcal{L}(e, \mu(\theta)). \quad (10)$$

The distributions among cross-domain classes will be enforced to be intra-class compact as well as inter-class separation by minimizing this objective, leading to class-level alignment.

4.3. Optimization of the network

The overall training loss is described as Eq. (11), which combines the supervised loss in Eq. (2), domain-level alignment loss in Eq. (3) and class-level alignment loss in Eq. (10).

$$\mathcal{L} = \mathcal{L}_{supervise} + \lambda_1 \mathcal{L}_{dom_adv} + \lambda_2 \mathcal{L}_{class}. \quad (11)$$

Where λ_1 and λ_2 are balanced parameters.

Since the optimization of class-level alignment loss needs to the pseudo-labels, we divide the training process into alternate two steps,

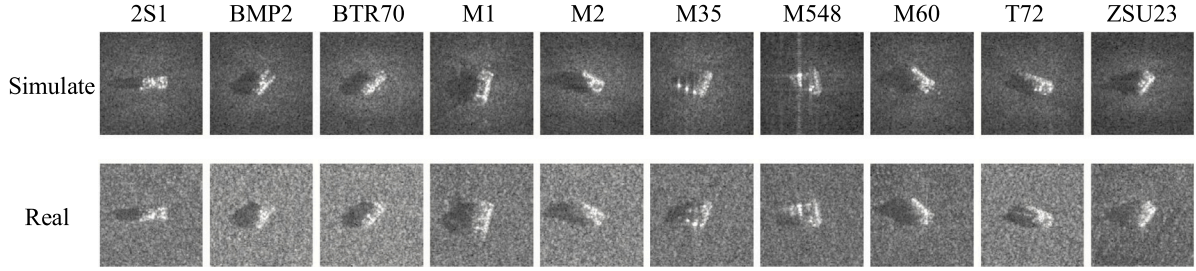


Fig. 4. Instances of a single image representing each class in the SAMPLE dataset.

Table 1

The Detailed information of Ten-Target data in the SAMPLE dataset.

Dataset	2s1	BMP2	BTR70	M1	M2	M35	M548	M60	T72	ZSU23
Simulate	174	107	92	129	128	129	128	176	108	174
Real-train	116	55	43	78	75	76	75	116	56	116
Real-test	58	52	49	51	53	53	53	60	52	58

Table 2

The detailed information of Five-Target data in the S2M dataset.

Dataset	2s1	BMP2	BTR70	T72	ZSU23
Simulate	174	107	92	108	174
Real-train	70	64	55	61	70
Real-test	59	43	50	37	65

i.e., self-training manner: (1) Fixing the parameters of network and prototypes, and updating the target pseudo-labels; (2) Based on the updated labels, updating the parameters of network and prototypes through back-propagation. These two steps are alternated in each training iteration. The supervised classification loss is minimized with labeled samples through stochastic gradient descent. The optimization of the domain-level alignment loss does not require labels, and also uses stochastic gradient descent to minimize the objective.

5. Experiments

5.1. Data description and implement details

We evaluate our method on two SAR target classification datasets, each of which includes a simulated and a real set to verify the superiority of the proposed method. The first dataset is the public SAMPLE (Lewis et al., 2019) dataset, which contains 10 classes of simulated/real vehicle targets. Table 1 gives the number of simulated and real SAR data of each class and Fig. 4 displays instances of the ten classes. The data are in elevation from 14° to 17° . In our settings, the elevation angle of test data is 17° and the elevation angles of training data are 14° , 15° , and 16° , respectively. The range azimuth angles of SAMPLE data are from 10° to 80° .

The second dataset is a self-build dataset (S2M dataset) collected from the SAMPLE and MSTAR (DAPRA and AFRL, 0000) datasets, which contains 5 classes of simulated/real vehicle targets. Specifically, we collected the data of the same category as the SAMPLE dataset in the MSTAR as the real set of the self-built dataset, where the azimuth angles are limited to 0° to 90° . Table 2 gives the number of simulated and real SAR data of each class. The simulated data are in elevation from 14° to 17° and the real data are in elevation 15° and 17° . In our settings, the elevation angle of real test data is 17° and the elevation angle of training data is 15° . For a given CAD model, the generation of simulated SAR data is mainly divided into two steps: 1) based on electromagnetic calculation, the electromagnetic scattering

characteristics of the target are predicted to generate radar echoes. 2) based on SAR imaging algorithm, radar echoes are imaged to generate complex images. In SAMPLE, the SAR imaging algorithm is the polar format algorithm. More details can be found at Lewis et al. (2019).

In our experimental setup, the source domain is the simulated SAR image set for which class labels are available. The target domain is the real SAR image set whose labels are not available. We follow the stipulations of inductive learning, where the target domain is split into training and testing sets. The training set and the source domain data participate in the learning of the UDA model together.

Next, the implementation details will be presented. In Eq. (11), we set the balanced parameters λ_1 and λ_2 as 0.5 and 1 respectively. The learning rate of our method is 0.001, and we set the decay as 0.1. The feature extractor in our method is ResNet50. The batch size is set to 64 and the network trains 15000 iterations. Our method is implemented with the Pytorch framework.

5.2. Evaluation metric

The probability of correct classification (PCC), which measures the proportion of correctly identified targets to the total number of targets, is a common metric used to assess the overall accuracy of classification.

In addition to accuracy, we use the Macro F1 score as a metric to evaluate the recognition performance, which is an evaluation metric that comprehensively considers the precision and recall of each category. The calculation formula is as follows:

$$Macro\ F1 = \frac{2 \times Macro\ Precision \times Macro\ Recall}{Macro\ Precision + Macro\ Recall}, \quad (12)$$

$$Macro\ Precision = \frac{1}{n} \sum_{i=1}^n Precision_i, \quad (13)$$

$$Macro\ Recall = \frac{1}{n} \sum_{i=1}^n Recall_i, \quad (14)$$

$$Precision_i = \frac{TP_i}{TP_i + FP_i}, i = 1, 2, \dots, n, \quad (15)$$

$$Recall_i = \frac{TP_i}{TP_i + FN_i}, i = 1, 2, \dots, n. \quad (16)$$

Where *Macro Precision* represents the average precision rate of each category, and *Macro Recall* represents the average recall rate of each category, TP_i represents the number of samples of class i classified as class i , FP_i represents the number of samples of other classes judged as class i (that is, false alarm), and FN_i represents the number of samples of class i judged as other classes (that is, missed detection).

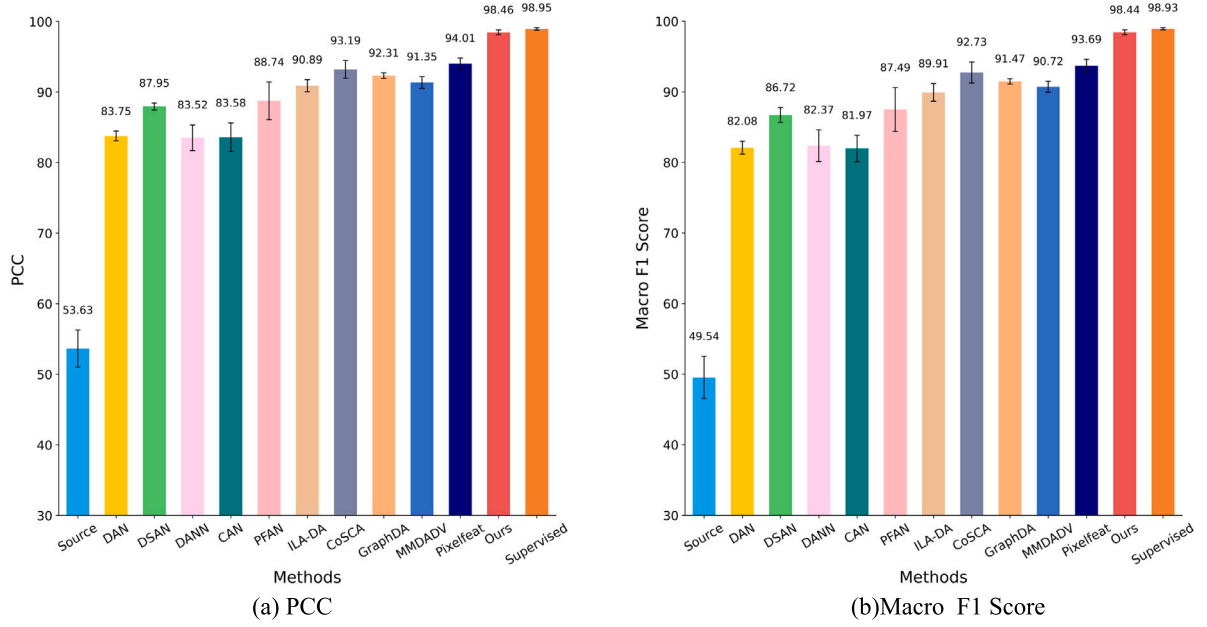


Fig. 5. The comparison results of different domain adaptive methods on the SAMPLE dataset.

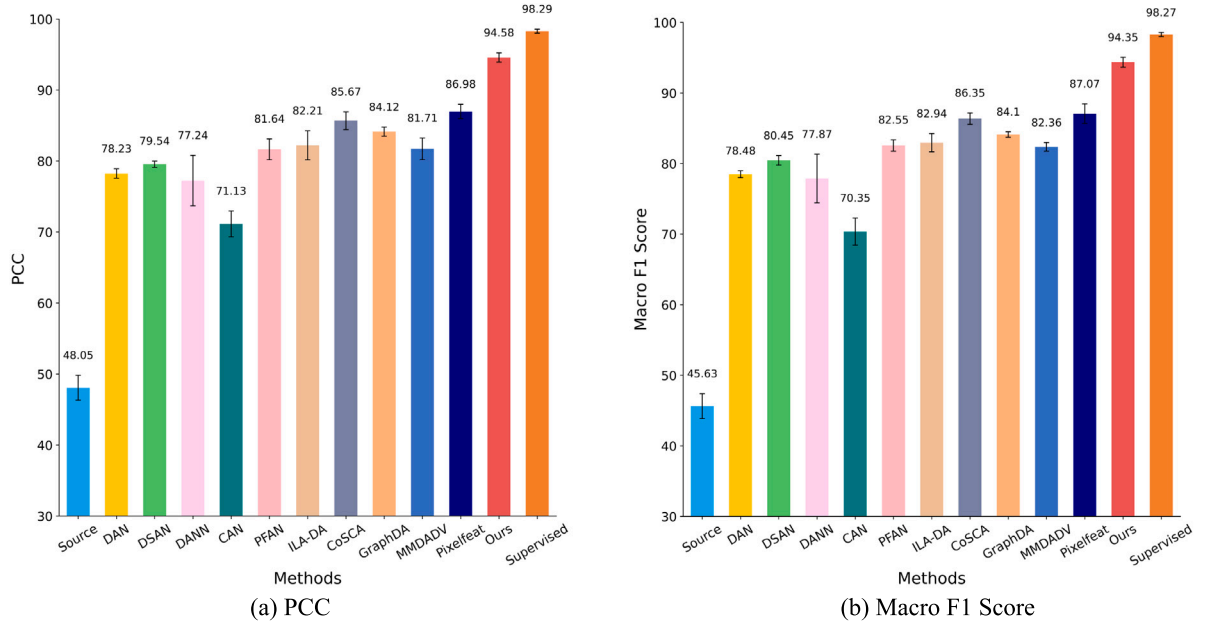


Fig. 6. The comparison results of different domain adaptive methods on the S2M dataset.

5.3. Performance comparison

(1) Compared with other methods

The effectiveness of our approach will be evaluated by comparing it with domain-level alignment UDA methods in computer vision and SAR target classification, as well as typical class-level alignment UDA methods in computer vision. Fig. 5 shows the overall and each class comparison results of the SAMPLE dataset. The results show the mean and standard deviation of five times. In Fig. 5, “Source only” denotes that directly perform the target classification model trained on the simulated (source) domain without domain adaptation to classify the test

samples of the real (target) domain, “DAN” denotes the domain-level alignment UDA methods of computer vision filed with MMD constraints in Long et al. (2015), “DANN” denotes the domain-level alignment UDA methods of computer vision filed with adversarial learning strategy in Ganin and Lempitsky (2015), “MMDADV” and “Pixelfeat” denote the domain-level alignment UDA methods of SAR target classification filed in Xiaoling et al. (2022), Chen et al. (2022b), “GrapgDA” (Saha et al., 2022) denotes the co-teaching-based method in remote sensing field, “DSAN”, “PFAN”, “CAN”, “ILA-DA” and “CoSCA” denote the class-level alignment UDA methods of computer vision filed in Zhu et al. (2020), Chen et al. (2019), Kang et al. (2019), Sharma et al. (2021), Dai et al. (2020).

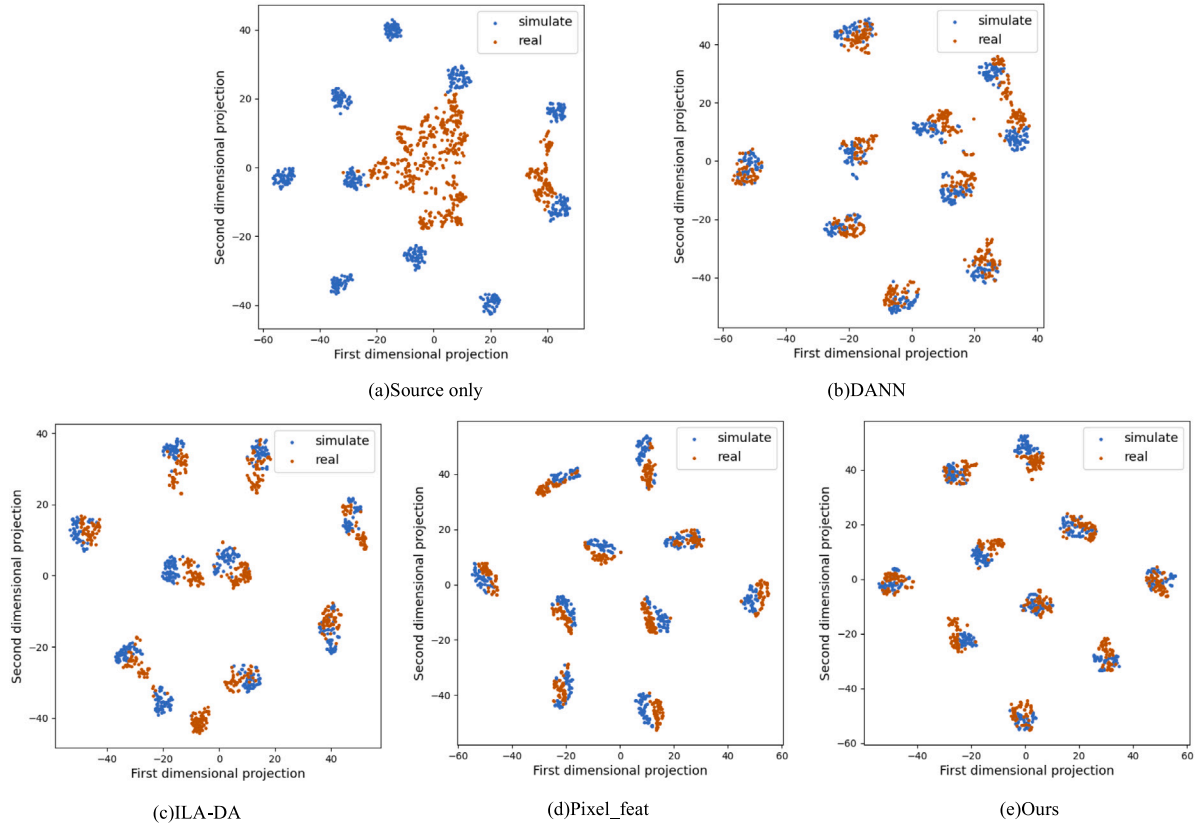


Fig. 7. The t-SNE feature visualization of cross-domain alignment for the classifier learned by “Source-only”, “DANN”, “ILA-DA”, “Pixel_feat”, and our method on the SAMPLE dataset. Red points denote features of the real (target) samples and blue points denote features of the simulated (source) samples.

The classification performance of “Source-only” without adaptation can be considered as a baseline. All the domain adaptive methods have demonstrated efficacy in enhancing performance compared with “Source only”, which shows the advantage of aligning the distributions between the two domains. The domain-level alignment UDA methods in computer vision (“DAN”, “DANN”) achieve limited performance boosting since they do not consider the fine-grained class structure. “MMDADV” and “Pixelfeat” are two UDA methods designed for SAR target classification, which achieve further performance improvement. “DSAN” and “CAN” are two class-level alignment UDA methods based on the metric constraint in computer vision. Due to lacking an insufficient filtering strategy for noisy pseudo labels, their performances are not satisfactory. “ILA-DA” alleviates the impact of wrong pseudo-labels to some extent through adversarial learning and multi-sample contrastive loss. GraphDA achieves better performance than most other class alignment methods in computer vision due to the co-teaching strategy to obtain more reliable pseudo-labels. CoSCA uses two MPL classifiers to get more reliable pseudo-labels and perform cross-domain contrastive learning to achieve better results. Since our method aligns the domains at domain-, and class-levels simultaneously, and effectively filters noisy pseudo labels, our method achieves the best classification result compared with all comparison methods, demonstrating the advancement of our method. It is noted that the proposed method achieves performance comparable to the fully supervised classification method (which can be regarded as an upper bound).

Fig. 6 shows the PCC and Macro F1 score performance over 5 classes on S2M. Since the simulation and real set have different angle ranges, S2M is more challenging than the SAMPLE dataset. Our method outperforms the other domain adaptation methods. But there is still a certain gap in the performance of the supervised method. It is noted that, since the simulated and real sets have large discrepancies and there is not sufficient pseudo-labels filtering strategy, class-level methods of the computer vision field are not competitive.

2) Visualization

• T-SNE visualization of cross-domain alignment

Fig. 7 (a)–(e) give the t-SNE feature visualization of cross-alignment in test split for the classifier learned by “Source-only”, “DANN”, “ILA-DA”, “Pixel_feat”, and our method on the SAMPLE dataset. Fig. 7 (a) illustrates that the distribution of real domain features is excursive and has significant domain discrepancy with simulated domain features for “Source-only”. The incorporation of the domain adaptation term through the “DANN”, “ILA-DA”, and “Pixel_feat” methods effectively reduces the distance between the source and target distributions. Since our method aligns the domains at domain-, and class-levels simultaneously, our method aligns the distributions more compactly, making the distribution of the target domain indistinguishable from the source domain.

• Confusion metric and feature t-SNE

Fig. 8 (a)–(e) and Fig. 9 (a)–(e) show the visualizations of the confusion MSTAR and feature t-SNE for the classifier learned by “Source-only”, “DANN”, “ILA-DA”, “Pixel_feat”, and our method. Through the confusion MSTAR and feature t-SNE of “Source-only”, we can see that the real and simulated domains have a large domain shift, making the classifier trained on simulated data cannot perform well on real data. The utilization of “DANN”, “ILA-DA”, and “Pixel_feat” for domain adaptation significantly reduces class confusion for the majority of classes. Notably, our method attains higher accuracy rates for all ten classes, indicating that the features learned by our approach are more distinguishing for the target domain. From Fig. 9 we can see that our method achieves better results for intra-class compact as well as inter-class separation.

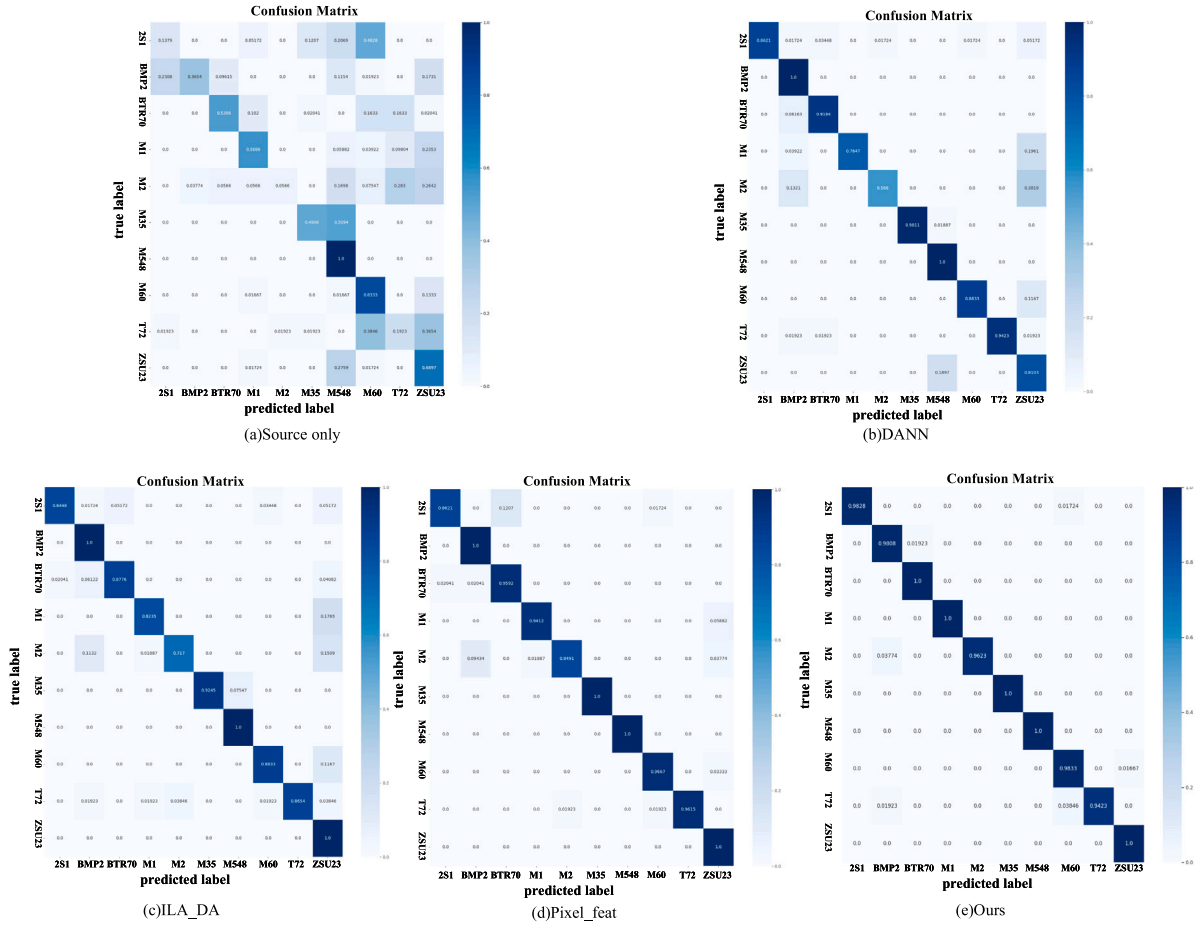


Fig. 8. The visualization of the confusion MSTAR for the “Source-only”, “DANN”, “ILA-DA”, “Pixel_feat”, and our method on the SAMPLE dataset. The MSTAR is organized such that each row represents the true category while each column corresponds to the predicted category.

Table 3
Ablation study.

Methods	Domain-Level	Class-Level	ASC	PCC	Macro F1
Source only	–	–	–	49.54 ± 0.980	53.63 ± 0.628
Ours	✓	–	–	90.20 ± 0.858	90.96 ± 0.683
	✓	✓	–	93.11 ± 0.464	93.58 ± 0.302
	–	✓	✓	87.95 ± 0.558	89.11 ± 0.656
	✓	✓	✓	95.77 ± 0.686	95.92 ± 0.664
	✓	✓	✓	98.44 ± 0.346	98.46 ± 0.337
Oracle	–	–	–	98.93 ± 0.153	98.95 ± 0.170

Table 4
The impact of weighted adversarial learning.

Methods	GRL	WGRL	PCC	F1 score
DANN	✓	–	86.72 ± 1.046	87.95 ± 0.494
	–	✓	90.20 ± 0.858	90.96 ± 0.683
Ours	✓	–	97.35 ± 0.328	97.43 ± 0.311
	–	✓	98.44 ± 0.345	98.46 ± 0.337

5.4. Model analysis

(1) Ablation Study

Ablation studies are always employed to better understand how the different components affect the behavior of the network. To comprehensively assess the impact of different levels on the performance of SAR target classification, we perform an ablation study on the SAMPLE dataset. Table 3 shows the ablation study results, where

Domain-level, Class-level, and ASC represent whether to use the proposed domain-level alignment, prototype-based class-level alignment, and pseudo label filtering combined with ASC, respectively. It is evident that all of the proposed modules are designed in a rational manner and that each module is capable of enhancing performance. Compared with “source only” without domain adaptation, the proposed gradient-weighted adversarial learning can effectively improve the classification performance (from 49.54% to 90.20% in terms of PCC and from 53.63% to 90.96% in terms of F1 score). Considering the class-level information in the distribution alignment can further achieve performance boosting. However, since the network training is affected by noisy pseudo labels, the performance of class-level alignment is still limited. Combining ASC information to filter the noisy pseudo labels can significantly reduce the influence of wrong pseudo labels and improve the classification performance. Using ASC only can be seen as a semi-supervised method with self-training. The guaranteed result of “ASC only” (from 49.54% to 87.95% in terms of PCC and from 53.63% to 89.11% in terms of F1 score) compared with “Source only”) can show the effectiveness

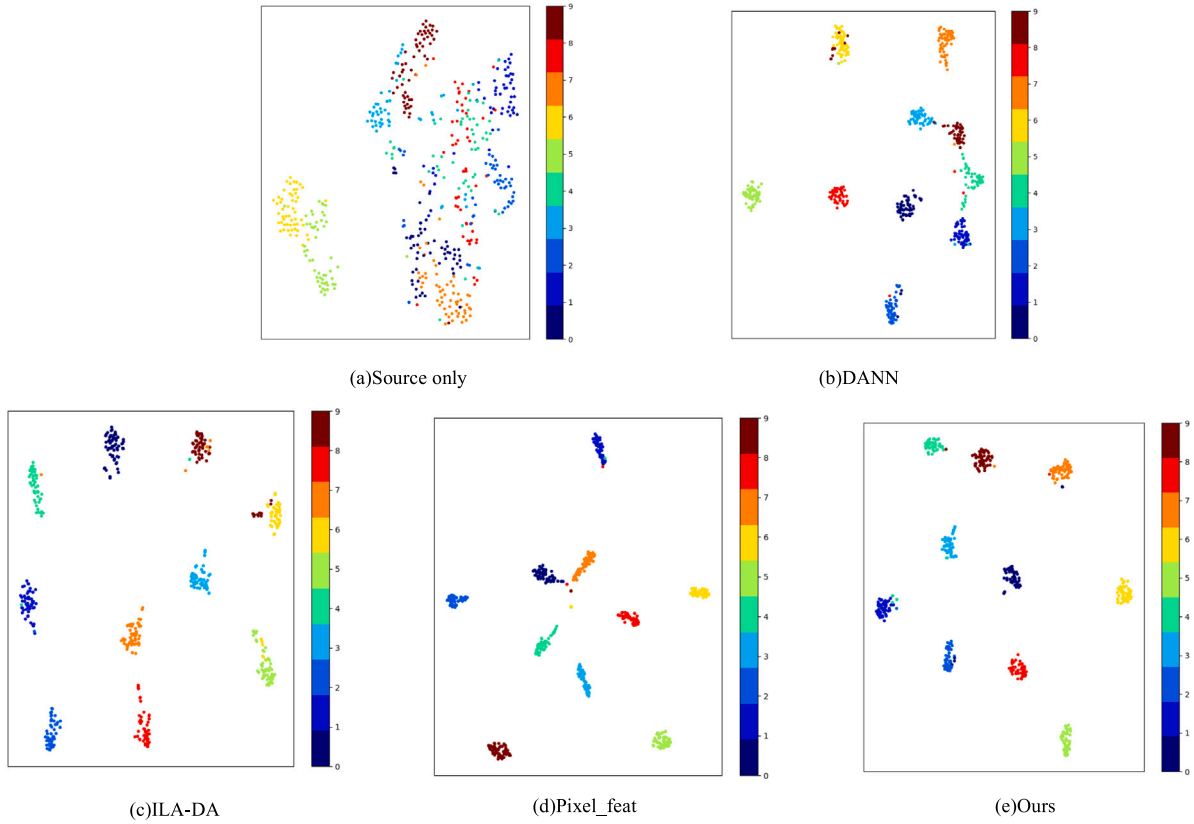


Fig. 9. The visualization of feature t-SNE for the classifier learned by “Source-only”, “DANN”, “ILA-DA”, “Pixel_feat”, and our method on the SAMPLE dataset. Different colors represent different classes. The number in the right column represents the ten classes, best viewed in colors. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of pseudo-label filtering. Compared with “Class-level alignment only”, combined ASC and prototype-based class-level alignment make the network capture the class structure of target-domain data distributions accurately, which improves the classification performance (from 87.95% to 95.77% in terms of PCC and from 89.11% to 95.92% in terms of F1 score). Furthermore, simultaneously using proposed domain-level alignment, prototype-based class-level alignment, and pseudo-label filtering combined with ASC significantly boosts the performance for overall cases and achieves the best classification result.

2) Analyzation of domain-level alignment

To verify the effectiveness of the proposed gradient-weighted adversarial learning, we compare the results of replacing the gradient reverse layer(GRL) module with Weighted GRL(WGRL) in vanilla DANN and our method, respectively. As shown in Table 4, our proposed WGRL achieves 3.48%, 1.09% improvement in terms of PCC and 3.01%, 1.03% improvement in terms of F1 score over GRL in vanilla DANN and our method, respectively, which demonstrates the effectiveness of our method.

To further verify that adversarial-based domain-level alignment can effectively confuse features, we show the changes in feature maps of the source and target domain before and after domain adaptation in Fig. 9. Since the low-level feature maps can better reflect the outline and structure information of the target, and can more intuitively reflect the differences between feature maps while the high-level feature maps are more abstract, we choose to visualize the low-level feature maps. Fig. 10 (a) shows that the feature maps before domain adaptation have a significant difference between the two domains. The feature maps of simulated SAR images are clean and noiseless while the feature maps of real SAR images are very noisy. After domain adaptation, the

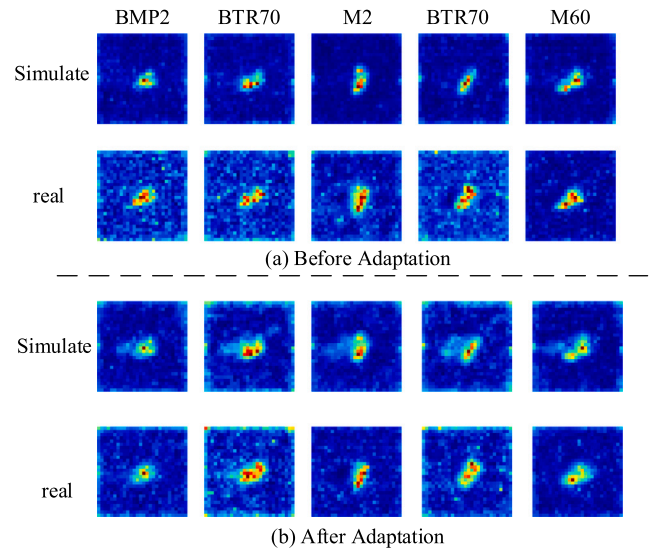


Fig. 10. The low-level feature maps of source and target domain before and after domain adaptation. The feature maps of the corresponding column in the first and third rows were generated from the same samples before and after domain adaptation, and similarly for the second and fourth rows. (a) Before domain adaptation. (b) After domain adaptation.

feature maps of the two domains are confused. Intuitively, the low-level features of the simulated domain tend to add noise while the low-level features of the real domain tend to denoise comparing the second and

Table 5

The comparison results of noisy pseudo labels filtering on SAMPLE dataset.

Strategy	Selected samples	Correctly predicted samples	Correct rate
CNN (threshold = 0.9)	406	291	0.7167
CNN combined with ASC	379	358	0.9445

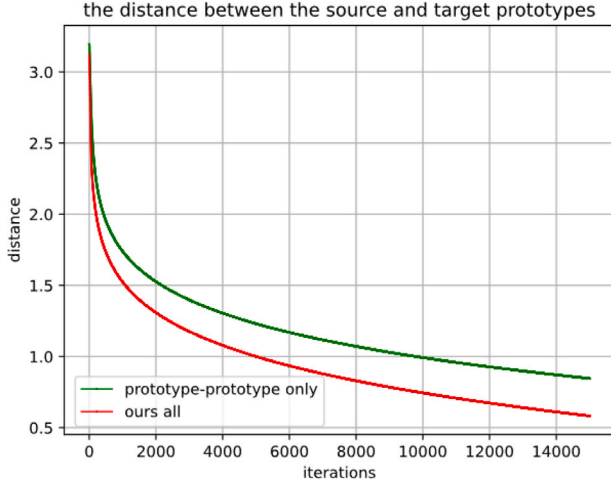


Fig. 11. The source and target prototypes distance with the increase of the iteration on the SAMPLE dataset. The green line represents only using cross-domain prototype-prototype modeling constraint and the red line represents using cross-domain prototype-prototype modeling constraint and using cross-domain prototype-instance modeling constraint simultaneously.

fourth rows. The confused features are further passed to the deep layers of the network and achieve better domain alignment.

3) Analysis of class-level alignment

The performance of the class-level alignment method is generally related to the quality of the predicted pseudo-labels, which may be noisy and lead to negative alignment between different classes if there are not sufficient filtering measures. To further show the effectiveness of our pseudo labels filtering strategy, we compare our method and traditional filtering solution on the SAMPLE dataset that only uses source-trained CNN with a confident threshold in Table 5. As shown in Table 5, though the traditional filtering solution can select more target samples to train the domain adaptive model, it will generate more wrong pseudo labels, which is harmful to the classifier training. Our method achieves better filtering performance and selects more correct pseudo labels, which ensures the correct guidelines of class-level alignment.

In Fig. 11, we visualize the source and target prototypes distance with the increase of the iteration, where the green line represents only using cross-domain prototype-prototype modeling constraint and the red line represents using cross-domain prototype-prototype modeling constraint and using cross-domain prototype-instance modeling constraint simultaneously. From Fig. 10 we can see that our cross-domain prototype-prototype modeling constraint can effectively pull closer the prototypes distance between source and target domains with the increase of the iteration. Since the relations of prototype and instance are learned with contrastive learning, it achieves more compact by using the cross-domain prototype-prototype modeling constraint and cross-domain prototype-instance modeling constraint simultaneously.

4) Analysis of λ_1 and λ_2

In this section, we analyze the impact of different values of hyperparameters in Eq. (11) on performance. The results are given in Fig. 1,

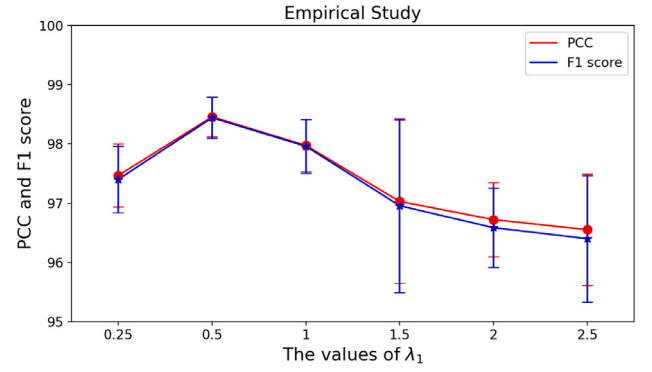
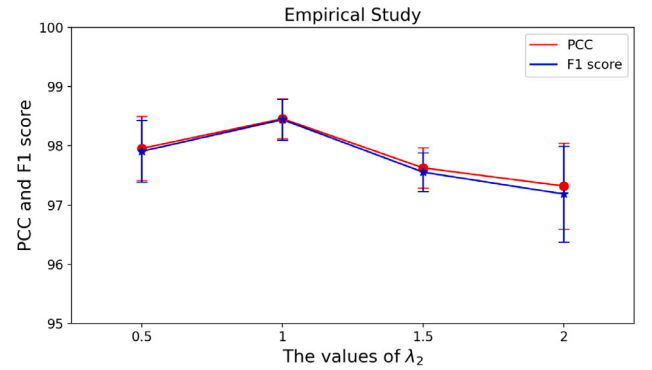
(a) λ_1 (b) λ_2

Fig. 12. The empirical study. (a) adjusting the value of λ_1 when we fix the value of λ_2 to 1. (b) adjusting the value of λ_2 when we fix the value of λ_1 to 0.5.

which show the mean and standard deviation of five times. Fig. 12 (a) shows the PCC and F1 score results of adjusting the value of λ_1 when we fix the value of λ_2 to 1. Fig. 12 (b) shows the PCC and F1 score results of adjusting the value of λ_2 when we fix the value of λ_1 to 0.5. From Fig. 12 we can see that when the value of λ_1 is set as 0.5 and the value of λ_2 is set as 1, the performance is best. The performance of the algorithm is not particularly sensitive to the settings of the hyperparameter values.

6. Conclusion

To solve the problem of lacking labeled data for training the SAR target classification network, we proposed a novel UDA method for SAR target classification by aligning the distribution between simulated and unlabeled real SAR domains across different granularity perspectives including domain-, and class-levels simultaneously. We first introduce the idea of class-level alignment into the SAR target classification task and propose a novel class-level alignment method based on the prototypical network. The cross-domain prototype-prototype relations and cross-domain prototype-instance relations are respectively modeled with contrastive learning to achieve intra-class prototype compact as well as inter-class prototype separation among cross-domain categories. From the view of the application, to solve the problem of severe unreliability of initial pseudo labels in the field of SAR target classification by class alignment method, a noisy pseudo label filtering strategy combined with the attribute scattering center is further proposed, which effectively improves the quality of the initial pseudo labels. Experiments conducted on the transfers across the SAMPLE and S2M dataset show the proposed method achieves remarkable unsupervised target classification performance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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